

Lab 3.1: Implementing Common Protocols and Services

Remove Lab Requirement Instructions in your submission

Lab Requirements:

1. In this assignment you will complete the assigned labs according to the instructions. This document will serve as a template to fill in answers where noted and screenshots required for each question.
2. Please make sure your screenshots follow the requirements for the course. You do not want to lose points needlessly for improper screenshots or formatting.
3. **Please make sure you show the syntax in your screenshots!** In order to get credit for your screenshots, you must show syntax and the output.
4. It is expected that any and all external sources used to complete any labs or support answers are properly cited in APA format and in a References Section at the end of the lab document.
5. Please name the file Lastname_LabTitle_Week# using your own last name at the beginning of the filename, lab title, and the appropriate week number. For example, if I were to name this lab: **Allen_Lab2.1_Week2.pdf**
6. Submit your work to the appropriate Assignment/Activity link in Blackboard. Do not submit assignments via course messages.
7. Points are assigned based on the completeness of your answers. Points will be deducted for errors and omissions.
8. There will be challenge flags throughout these labs. These are not required to be completed for the assignments, but feel free to complete them on your own. Also feel free

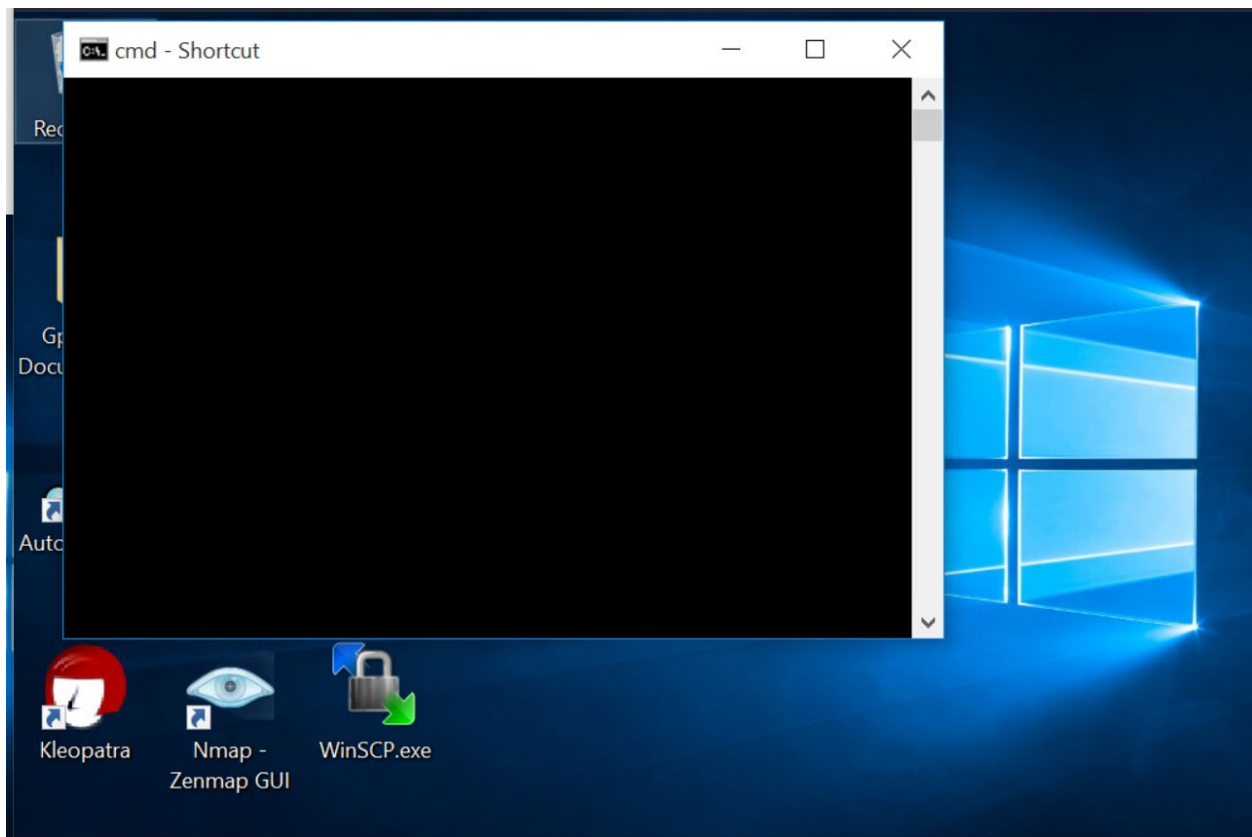
to go through the labs as many times as you want. Going through them a couple of times can help students understand the material.

Lab 3.1: Part 1 – Using Telnet and SSH Protocols

1. Describe what the `sc query tlntsvr` command does

Replace with written response

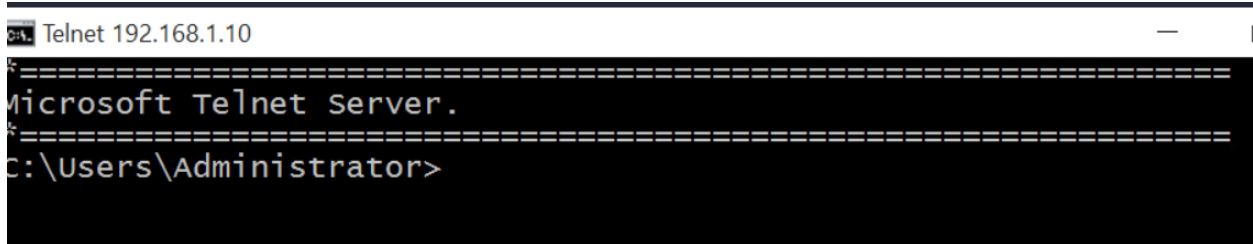
2. Question 5



3. Describe what we are doing with the **nmap** command in question 5

We are verifying that the telnet services are open in the windows server by using nmap

4. Question 9

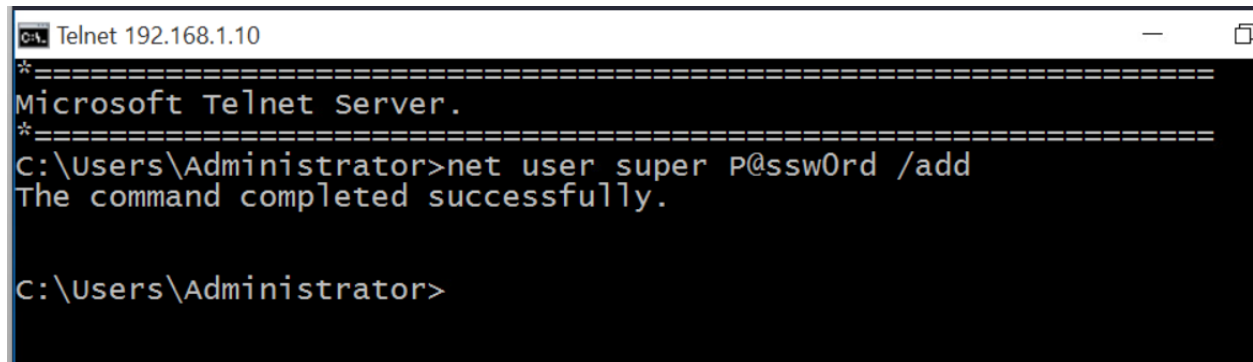


```
CHL Telnet 192.168.1.10
*=====
Microsoft Telnet Server.
*=====
C:\Users\Administrator>
```

5. Describe what we are doing in question 9

Logging into the telnet server on the windows server machine using its credentials.

6. Question 10



```
CHL Telnet 192.168.1.10
*=====
Microsoft Telnet Server.
*=====
C:\Users\Administrator>net user super P@ssw0rd /add
The command completed successfully.

C:\Users\Administrator>
```

7. What we are doing in question 10

Adding a user called super and changing the password for it.

8. Question 16

```
C:\>sc stop tlntsvr

SERVICE_NAME: tlntsvr
        TYPE               : 10  WIN32_OWN_PROCESS
        STATE                : 3   STOP_PENDING
                                (STOPPABLE, PAUSABLE, ACCEPTS_SHUTDOWN)
        WIN32_EXIT_CODE       : 0   (0x0)
        SERVICE_EXIT_CODE   : 0   (0x0)
        CHECKPOINT           : 0x0
        WAIT_HINT            : 0x0

C:\>
```

9. Describe what we are doing in question 16

Stopping the telnet server on the windows server machine

10. Question 23



11. Question 26

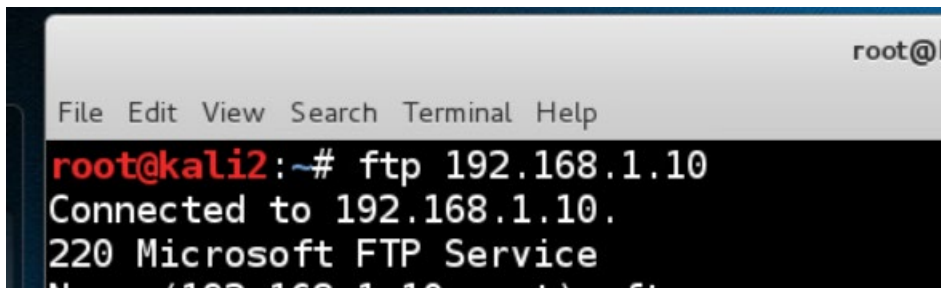
```
root@kali2:~# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
```

Lab 3.1: Part 2 – Using the FTP and SCP Protocols for File Transfer

1. Research and describe what the command **ftp** does

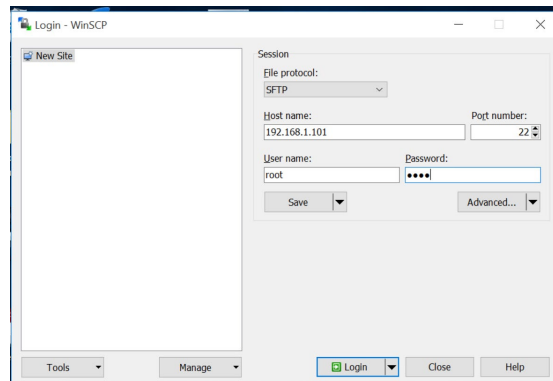
The ftp command is a standard network utility used to transfer files between a local computer and a remote server over the File Transfer Protocol.

2. Question 2

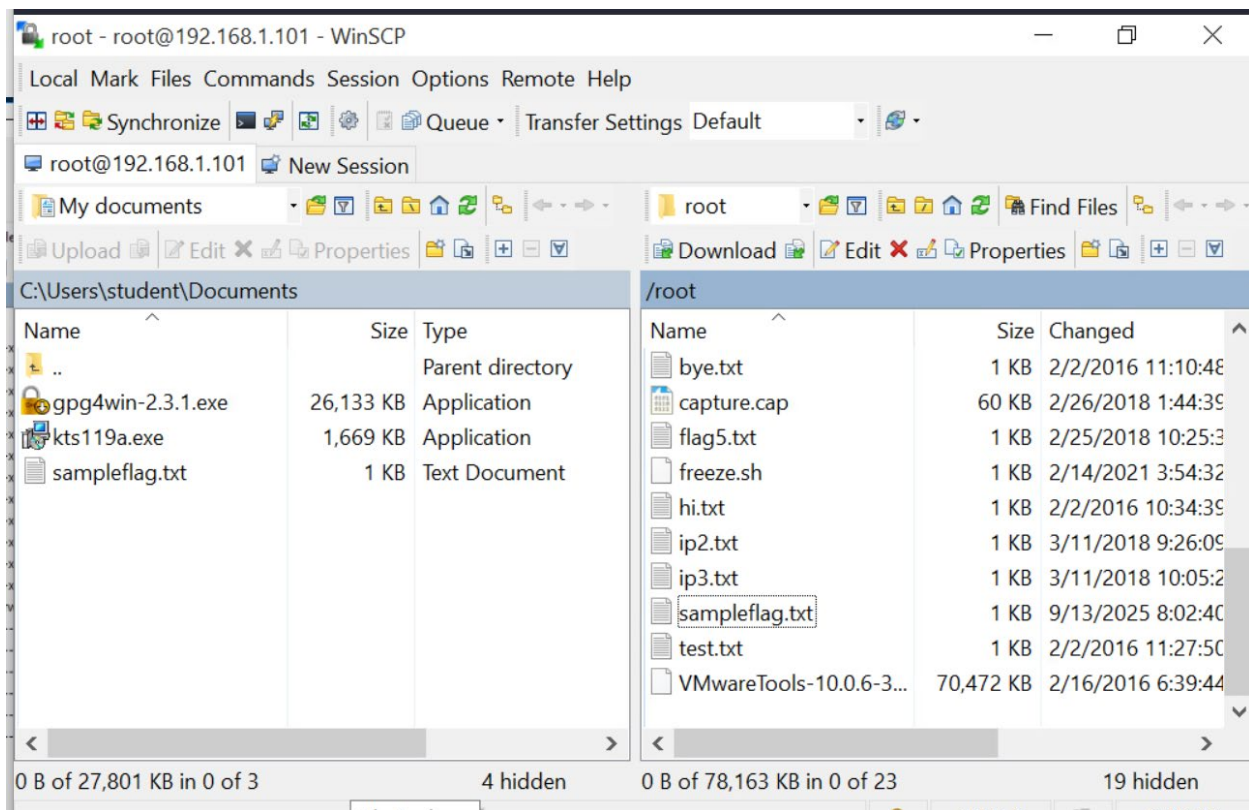


```
root@kali2:~# ftp 192.168.1.10
Connected to 192.168.1.10.
220 Microsoft FTP Service
```

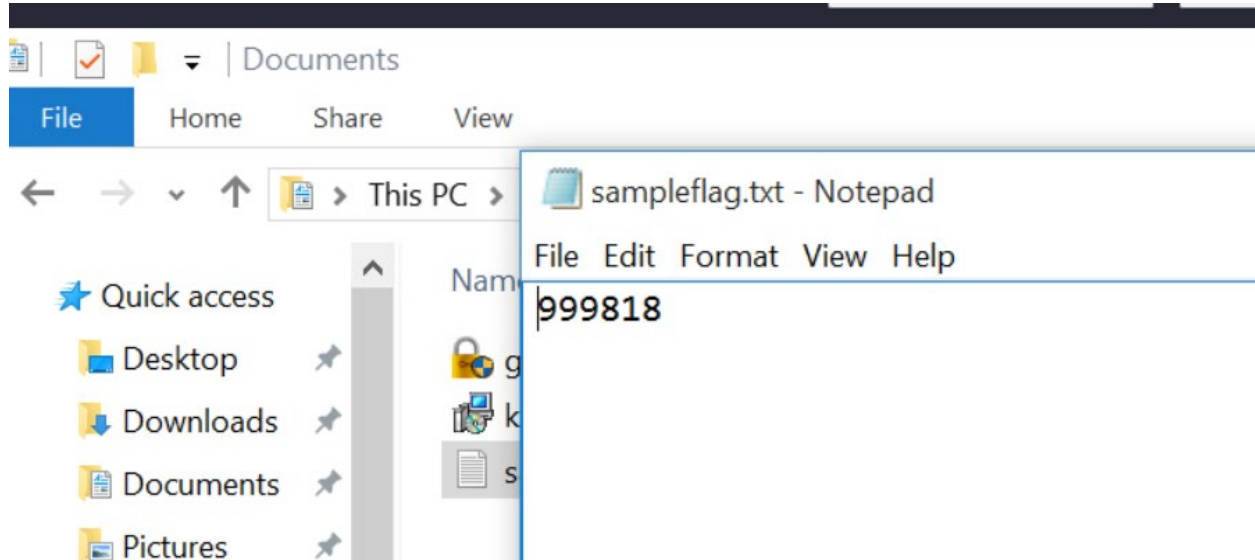
3. Question 10



4. Question 12



5. Question 17



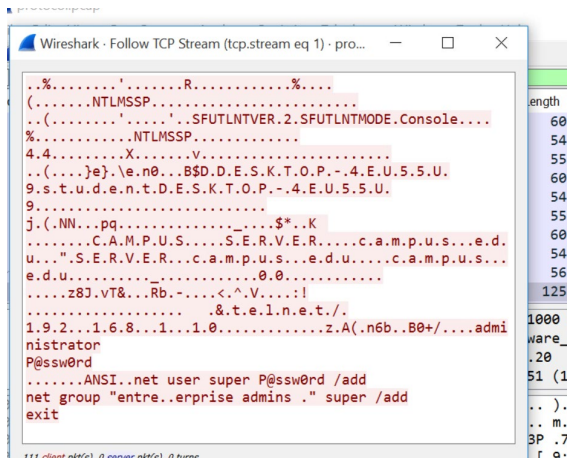
Lab 3.1: Part 3 – Examining Protocol Traffic in Wireshark

1. Question 3

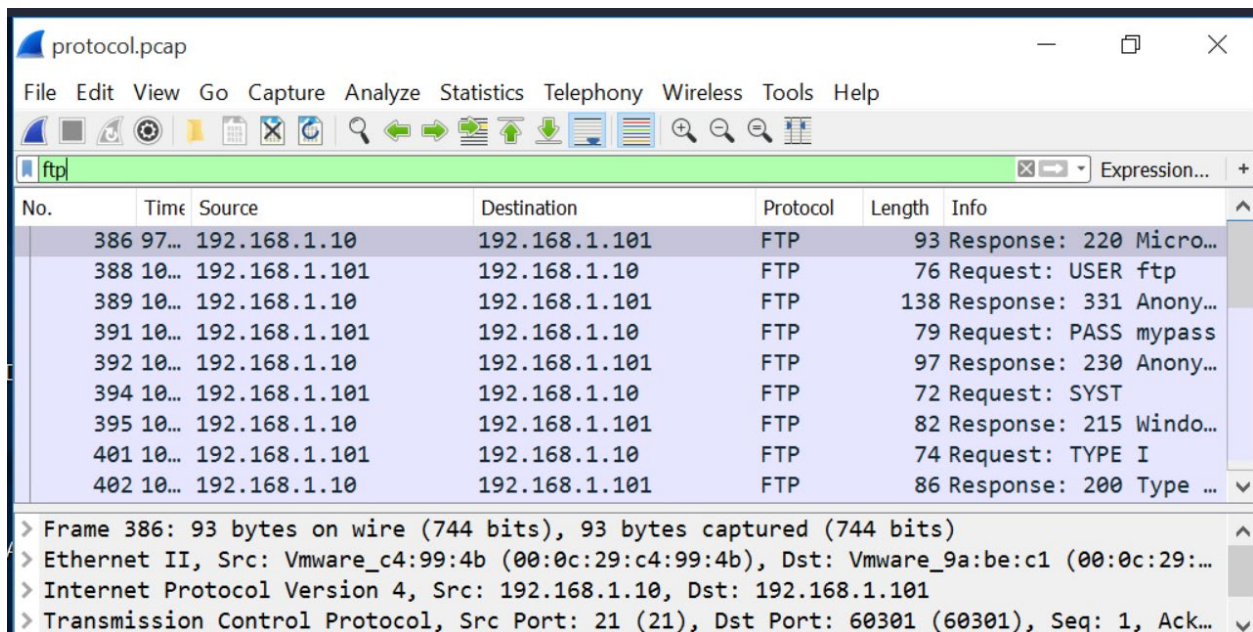
Before



After:



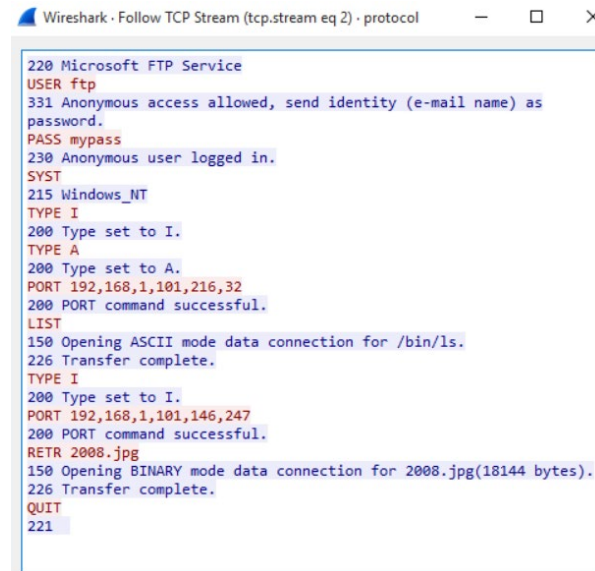
2. Question 5



3. What does the ftp filter in Wireshark do?

The ftp filter in Wireshark displays only the packets related to the File Transfer Protocol, such as FTP commands, responses, and file transfer control messages exchanged between client and server.

4. Question 7

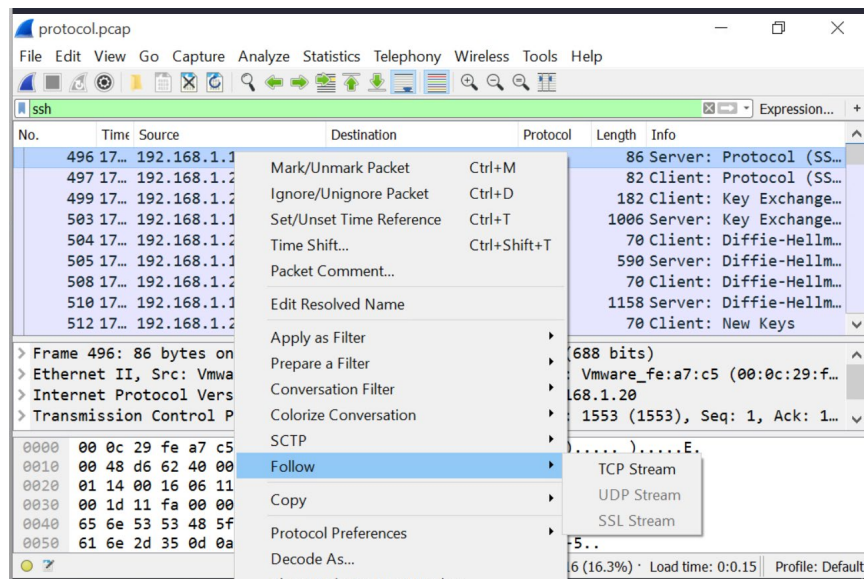


```
220 Microsoft FTP Service
USER ftp
331 Anonymous access allowed, send identity (e-mail name) as
password.
PASS mypass
230 Anonymous user logged in.
SYST
215 Windows_NT
TYPE I
200 Type set to I.
TYPE A
200 Type set to A.
PORT 192,168,1,101,216,32
200 PORT command successful.
LIST
150 Opening ASCII mode data connection for /bin/ls.
226 Transfer complete.
TYPE I
200 Type set to I.
PORT 192,168,1,101,146,247
200 PORT command successful.
RETR 2008.jpg
150 Opening BINARY mode data connection for 2008.jpg(18144 bytes).
226 Transfer complete.
QUIT
221
```

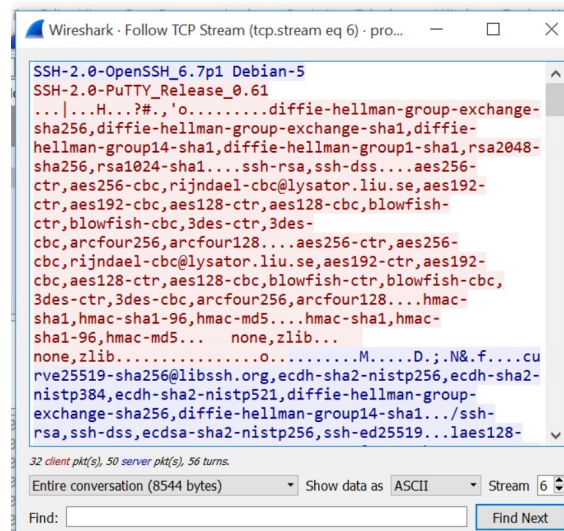
1. What does the **ssh** filter in **Wireshark** do?

The ssh filter in Wireshark shows only packets related to the Secure Shell (SSH) protocol, including encrypted login sessions, remote command execution, and secure file transfers over SSH.

2. Question 8



3. Question 10



Lab 3.1: Part 4 – Lessons Learned

Through this lab, I learned the importance of understanding and differentiating between various network protocols such as Telnet, SSH, FTP, and SCP. Each protocol serves a specific purpose in managing remote systems and transferring files, but they also differ in terms of security and efficiency. Another key takeaway was the practical use of tools like nmap, command-line utilities, and Wireshark. By running specific commands and analyzing traffic, I was able to verify open services, create and manage user accounts, and monitor network activity at the packet level. Finally, this lab emphasized the importance of stopping unnecessary services and properly managing user access. Leaving services like Telnet enabled can pose significant security risks if not monitored or restricted. By learning to disable services, configure secure alternatives, and track activity with the right tools, I gained a clearer understanding of how to maintain both functionality and security in a networked environment.